RODRIGO GOMES DE ARAÚJO

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Summary

MSc Computer Science student (graduating **July 2027**) focused on systems where **correctness and robustness** are key. **Co-founder and lead engineer** of Tinker, a dev tool that guarantees only **verified, reviewed and in-spec** code reaches production. Built a **proof-of-work blockchain** in Rust and is currently extending session types to **enforce correct usage of cryptographic resources**. Thrives in **fast-paced** and **high-stakes** environments and emphasizes **testing, documentation and API design** in the development process. Based in Porto, Portugal and **open to international relocation or remote work**.

Education

Master's — **Sep 2025 – Present (Jul 2027)**
Computer Science — *University of Porto (FCUP)*

Bachelor's — **Sep 2022 – Jul 2025**
Computer Science and Engineering — *University of Porto (FEUP)*

Experience

Tinker — **Jul 2026 – Present**
Co-founder & Lead Engineer — *Rust, Gitoxide, Ed25519, IPLD/DAG-CBOR, Crlu, Serde*

- Co-founded and lead a **3-person startup**. **Primary author of the codebase** and owner of the architecture, product direction and sprint planning.
- Tinker merges **version control, project management and CI** into one system.
- **Fits how teams already work**, from everyday development to critical releases or **AI agents implementing in parallel**. Each change gets the level of authority, verification and review it needs, and the platform enforces it. The project board tracks real progress on its own, with no manual updates.
- Designed the **security and trust model**. Signed approvals bound to the exact action, **tiered authority** separating agents from humans and a **tamper-evident history** linking every change in production to its spec, verification and approval.
- Fully built in **Rust** with a Git underlay and **concurrent verification** of immutable versions.

Research

CaST (Cryptographically-aware Session Types) — **Jul 2026 – Present (Jul 2027)**
Master's Thesis — *FreeST, Haskell, Type Systems, Applied Cryptography*

- Extending session types to capture **cryptographic resource usage**, so that its misuse in protocols is rejected at compile-time.
- Formalizing the extension and providing a **soundness proof**, integrating it into the FreeST compiler and exploring applications in smart contracts and protocol specification.

Coursework

PoW Blockchain for Decentralized Auctions
Data Systems Security · 19/20 — *Rust, Tokio, LibP2P, Docker, Blake2b, Ed25519*

- Extensive **research on the state of the art** of Bitcoin, Ethereum and Solana and investigation regarding the security and resilience to failure of these systems.
- Developed a DHT overlay based on Kademlia for **decentralized network construction**.
- PoW Blockchain with **Blake2b hashing**, **Ed25519 signatures** and **Merkle Trees** for transaction validation.
- **Client for auctions** using the blockchain layer as a public ledger.
- Docker **adversarial network simulation** with clients and bootstrap, well behaved and malicious nodes.

Portuguese Elections Analytics Platform | PostgreSQL, PostGIS, Python, Plotly, FastAPI, OCaml

- **Interactive data visualization dashboard** for 10 years of Portuguese elections. Choropleth vote-swing maps, treemaps, radar and multi-year trend charts, drillable from country down to parish.
- **Geospatial rendering in-database** with topology-preserving simplification adapted to zoom level.
- **ETL pipeline** into a **star-schema warehouse**, feeding **complex analytical queries** that join spatial and electoral data (recursive roll-ups, window functions, OLAP cubes).

Awards

2nd Place in IEEE RetroJam 2025 | IEEE UP Student Branch — **Oct 2025**

- Game from scratch in 48 hours (Rust and Raylib), learning how to **reduce a problem to its core requirements**.

Academic Merit Awards — **2020, 2021, 2022**

1st Place in the National Robotics Championship | Instituto Politécnico de Bragança — **May 2016**

Technical Skills

Languages: Rust, C/C++, Haskell, Python, SQL, Java
Concepts: Type Systems, Formal Methods, Applied Cryptography, Concurrency, Distributed Systems
Systems & Tools: Tokio, LibP2P, gRPC (Tonic), Git, UNIX/Linux, Docker, Terraform, OpenMP
Data: PostgreSQL, PostGIS, pandas, NumPy, scikit-learn, Plotly, Bonsai
Spoken Languages: Portuguese (Native), English (Fluent), Spanish (Intermediate)